

Functional Annotation Report: *bcsB* (Q88JL3, PP_2636) in *Pseudomonas putida* KT2440

Summary

The gene **bcsB** (UniProt Q88JL3; ordered locus PP_2636) of *Pseudomonas putida* KT2440 encodes **BcsB, the membrane-anchored, periplasmic accessory (regulatory) subunit of bacterial cellulose synthase**. BcsB is not itself a catalytic enzyme. Instead, it forms an obligate, stoichiometric inner-membrane complex with the catalytic glycosyltransferase **BcsA**, which uses **UDP-glucose** as the sugar-nucleotide donor to processively polymerize **β -1,4-glucan (cellulose)** one glucose at a time and simultaneously translocates the nascent polymer across the cytoplasmic (inner) membrane. Within this BcsA–BcsB "terminal complex," BcsB provides the periplasmic scaffold that receives the emerging glucan chain and guides it toward the outer-membrane secretion channel. This role is firmly established by high-resolution crystal structures of the homologous BcsA–BcsB complex from *Rhodobacter sphaeroides* and by biochemical reconstitution of the AcsAB/BcsAB complex from acetobacteria, and it is assigned to Q88JL3 through unambiguous membership in the **AcsB/BcsB protein family** (Pfam PF03170; InterPro IPR003920/IPR018513).

Functionally, the synthase is an **allosterically gated enzyme controlled by the bacterial second messenger cyclic di-GMP (c-di-GMP)**. c-di-GMP binding releases an autoinhibited state of BcsA by breaking a salt bridge that tethers a "gating loop" over the active site, switching the enzyme on. BcsB participates in the machine's regulation beyond catalysis: it acts as a **hub of a megadalton Bcs macrocomplex** and physically recruits c-di-GMP–metabolizing enzymes (a diguanylate cyclase and a phosphodiesterase) to establish a **local source and sink of c-di-GMP** immediately adjacent to the synthase. The product cellulose is secreted across the cell envelope through a trans-envelope pathway: BcsA channel → periplasm (BcsB) → the outer-membrane porin **BcsC**.

In the specific context of *P. putida* KT2440, cellulose produced by the Bcs system is a **water-stress-induced biofilm matrix component**. Experimental work in *P. putida* shows that cellulose (Bcs) plays a comparatively minor role in overall biofilm architecture (where a novel

exopolysaccharide, Pea, dominates) but contributes measurably to **desiccation tolerance and rhizosphere (root) colonization**, and that *bcs* expression is strongly induced by water/matrix stress. Thus, BcsB's molecular job — enabling cellulose synthesis and export — translates at the organismal level into ecological fitness under water-limiting conditions in the plant rhizosphere.

Key Findings

Finding 1 — BcsB is the membrane-anchored, periplasmic co-subunit of the cellulose-synthesizing/translocating complex

The primary identity of the *bcsB* gene product is structural/functional rather than catalytic. The definitive evidence comes from the crystal structure of the *Rhodobacter sphaeroides* BcsA–BcsB complex captured with a translocating polysaccharide, which showed that "**cellulose synthesis and transport across the inner bacterial membrane is mediated by a complex of the membrane-integrated catalytic BcsA subunit and the membrane-anchored, periplasmic BcsB protein**" (PMID: 23222542). The same study established the processive mechanism whereby "**the nascent polysaccharide is extended by one glucose molecule at a time,**" a single-glucose addition cycle in which BcsB participates in translocation and stabilization of the emerging chain.

The assignment of this role to *P. putida* Q88JL3 is secure by sequence and domain evidence: Q88JL3 belongs to the **AcsB/BcsB family** and carries the **BcsB Pfam domain (PF03170)** together with **Cellulose_synth_B** domains (**IPR003920 / IPR018513**), which are precisely the domains of the crystallographically characterized BcsB. In other words, the *P. putida* protein is the direct ortholog of the structurally defined BcsB.

Interpretation: BcsB is a periplasmic scaffold/adaptor, not an enzyme. Its "substrate" is the nascent β -1,4-glucan chain, which it receives from BcsA and guides through the periplasm. The membrane anchor tethers this periplasmic domain at the inner membrane immediately above the catalytic exit channel of BcsA.

Finding 2 — The BcsA–BcsB synthase is allosterically activated by c-di-GMP

The BcsA–BcsB complex is a **c-di-GMP-regulated enzyme**, which explains why UniProt annotates Q88JL3 as a "cyclic di-GMP-binding protein / cellulose synthase regulatory subunit." Crystal structures of the c-di-GMP-activated complex revealed the mechanism: "**c-di-GMP releases an autoinhibited state of the enzyme by breaking a salt bridge that otherwise tethers a conserved gating loop that controls access to and substrate coordination at the active site**" (PMID: 24704788). Disrupting this salt bridge genetically yields a constitutively active synthase, confirming the gating model. The broader physiological logic is that "**the bacterial signaling molecule cyclic di-GMP (c-di-GMP) stimulates the synthesis of bacterial cellulose, which is frequently found in biofilms**" (PMID: 24704788).

Interpretation: Cellulose production is switched on when intracellular c-di-GMP rises (the sessile/biofilm state). The primary c-di-GMP sensor is the PilZ domain of BcsA, but BcsB is an integral member of the regulated complex and — as Finding 3 shows — organizes the local signaling environment that feeds c-di-GMP to the synthase.

Finding 3 — BcsB is a hub of a megadalton Bcs macrocomplex and organizes local c-di-GMP signaling at the synthase

Rather than functioning only as a minimal BcsA–BcsB pair, the cellulose machinery assembles into a large multi-subunit device. Electron microscopy of the *E. coli* Bcs system revealed "**a stable, megadalton-sized macromolecular assembly, which encompasses most of the inner membrane and cytosolic Bcs components and features a previously unobserved asymmetric architecture**" (PMID: 29234007); accessory proteins in this assembly regulate secretion by modulating its assembly and stability.

Crucially, BcsB serves as a **scaffold for localized second-messenger signaling**. In *E. coli*, "**DgcC and a particular PDE, PdeK (encoded right next to the cellulose operon), directly interact with cellulose synthase subunit BcsB and with each other, thus establishing physical proximity between cellulose synthase and a local source and sink of c-di-GMP**" (PMID: 32534064). This creates a spatially confined c-di-GMP microdomain adjacent to BcsA, whose activation requires the allosteric ligand (Finding 2).

Interpretation: BcsB has a dual role — it is both a structural subunit of the synthase and a **signaling scaffold**. By recruiting a diguanylate cyclase (source) and a phosphodiesterase (sink), BcsB tunes c-di-GMP concentration precisely where it is needed, coupling the cell's signaling state to the on/off state of cellulose synthesis.

Finding 4 — In *P. putida* KT2440, the Bcs/cellulose system supports desiccation tolerance and rhizosphere colonization

Organism-specific functional data place BcsB's role in ecological context. In *P. putida* mt-2, "**Bcs plays a minor role in biofilm formation and stability, although it does contribute to rhizosphere colonization based on a competition assay**," while a novel exopolysaccharide (Pea) dominates biofilm architecture; importantly, "**both forms of water stress highly induced bcs expression**" (PMID: 21507177). The intracellular c-di-GMP pool that governs this planktonic-to-biofilm switch in *P. putida* KT2440 is set by a large complement of diguanylate cyclases and phosphodiesterases — for example the salt-responsive CfcA/CfcR two-component system (PMID: 35001503) and the LapD-interacting phosphodiesterase DibA/BifA network (PMID: 37927230).

Interpretation: In *P. putida*, cellulose is not the principal biofilm structural polymer, but it is a **stress-responsive matrix component** that promotes survival under water limitation and enhances competitive colonization of plant roots. This is the physiological output of BcsB's molecular function.

Finding 5 — The synthase uses UDP-glucose to make β -1,4-glucan, and the catalytic complex sits in the inner (cytoplasmic) membrane

Biochemical reconstitution confirms the reaction and localization. A "**highly purified AcsAB complex ... catalyzed incorporation of UDP-glucose into β -1,4-glucan chains, and responded to the presence of allosteric activator**" (c-di-GMP), and "**the catalytic AcsAB complex is embedded in the cytoplasmic membrane**" (PMID: 27214134). Because Q88JL3 belongs to the AcsB/BcsB family, *P. putida* BcsB is the AcsB-equivalent accessory subunit of this inner-membrane terminal complex.

Interpretation: This pins down the biochemistry precisely: **donor substrate = UDP-glucose; product = β -1,4-glucan (cellulose); location = inner/cytoplasmic membrane; regulation = allosteric c-di-GMP activation**. BcsB is the accessory subunit within this reconstituted, activity-competent complex.

Finding 6 — BcsB guides nascent cellulose through the periplasm toward the BcsC outer-membrane porin

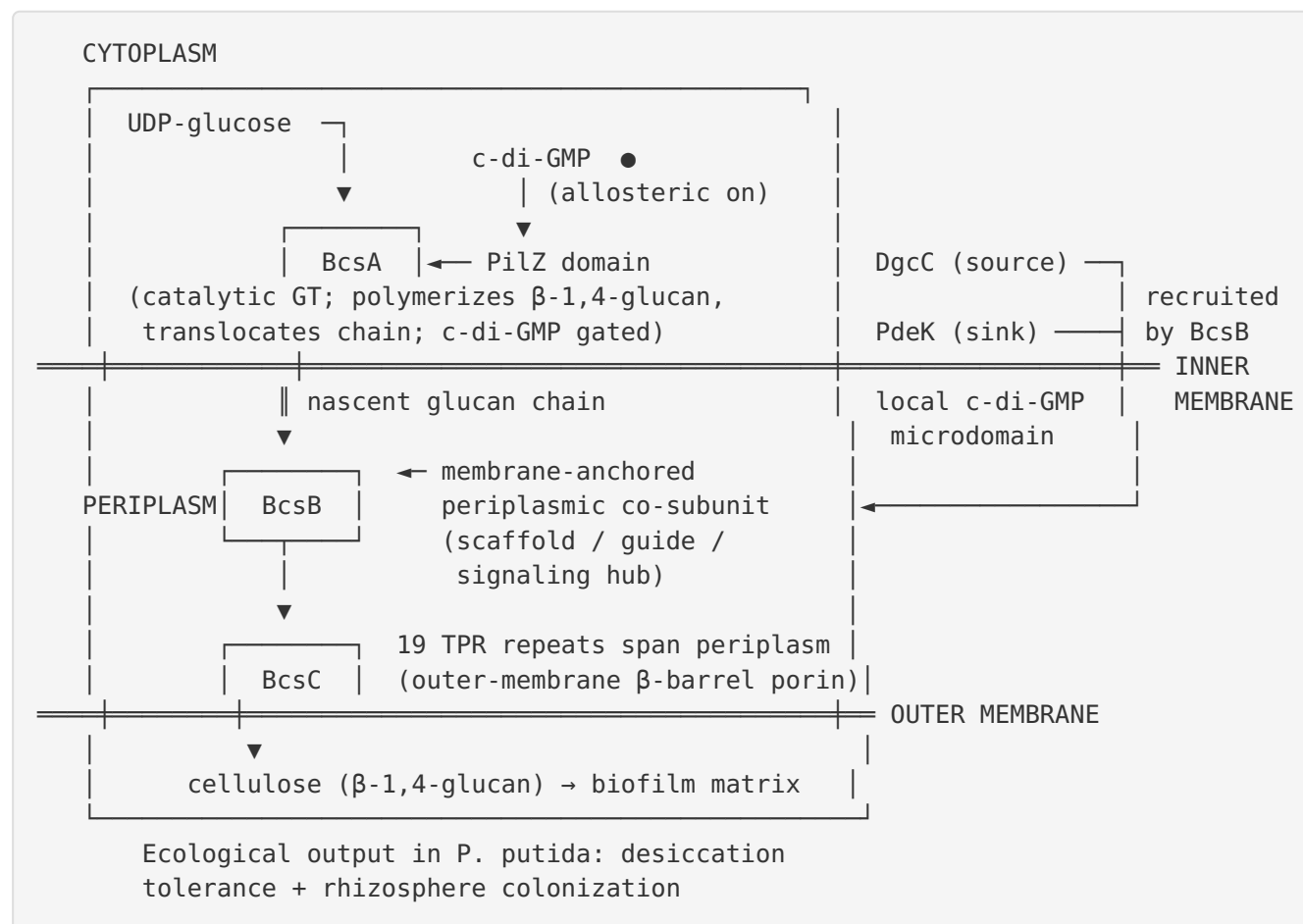
Cellulose export is a two-membrane journey. "**The BcsA subunit synthesizes cellulose and also transports the polymer across the inner membrane. Translocation across the outer membrane occurs through the BcsC porin, which extends into the periplasm via 19 tetra-**

tricopeptide repeats (TPR)" (PMID: 31604608). BcsC forms a 16-stranded β -barrel whose channel is lined with hydrophilic and aromatic residues, suggesting facilitated diffusion of cellulose via aromatic stacking and hydrogen bonding. The **membrane-anchored, periplasmic BcsB sits between the inner-membrane BcsA channel and the periplasm-spanning BcsC**, bridging the gap in the secretion path.

Interpretation: BcsB's periplasmic location is functionally essential: it occupies and helps organize the periplasmic space where the nascent glucan must travel from BcsA to BcsC, defining a continuous **trans-envelope secretion conduit**.

Mechanistic Model / Interpretation

Bringing the six findings together yields a coherent molecular model of BcsB's function in *P. putida* KT2440:



Step-by-step:

1. **Signal input.** Rising intracellular c-di-GMP (produced by diguanylate cyclases such as CfcR and, locally, by DgcC recruited to BcsB) reflects a shift toward the sessile/biofilm lifestyle.
2. **Allosteric activation.** c-di-GMP binds BcsA's PilZ domain, breaking a salt bridge and releasing the gating loop over the active site, switching on catalysis ([PMID: 24704788](#)).
3. **Polymerization.** BcsA uses UDP-glucose to add glucose to the growing chain one residue at a time, forming β -1,4-glucan ([PMID: 23222542](#); [PMID: 27214134](#)).
4. **Inner-membrane translocation.** The same BcsA channel threads the nascent glucan across the inner membrane into the periplasm.
5. **Periplasmic hand-off. BcsB**, anchored at the inner membrane with its bulk in the periplasm, receives and guides the chain and scaffolds the assembly ([PMID: 23222542](#); [PMID: 29234007](#)).
6. **Outer-membrane secretion.** BcsC's periplasmic TPR array meets the chain and its β -barrel exports cellulose to the cell surface ([PMID: 31604608](#)).
7. **Local signaling control.** BcsB's recruitment of DgcC/PdeK maintains a c-di-GMP source-and-sink microdomain at the synthase for responsive on/off control ([PMID: 32534064](#)).
8. **Ecological output.** Secreted cellulose contributes to the biofilm matrix and promotes desiccation tolerance and root colonization in *P. putida* ([PMID: 21507177](#)).

Summary table of BcsB's molecular attributes

Attribute	Assignment	Evidence
Molecular role	Periplasmic accessory/regulatory co-subunit of cellulose synthase (non-catalytic)	PMID: 23222542
Obligate partner	Catalytic glycosyltransferase BcsA	PMID: 23222542
Reaction catalyzed by the complex	UDP-glucose → β -1,4-glucan (cellulose) + UDP	PMID: 27214134
Donor substrate	UDP-glucose	PMID: 27214134
Product	β -1,4-glucan (cellulose)	PMID: 27214134
Localization	Inner/cytoplasmic membrane anchor; bulk domain in periplasm	PMID: 27214134 ; PMID: 23222542
Regulation	Allosteric activation of complex by c-di-GMP; BcsB scaffolds local c-di-GMP enzymes	PMID: 24704788 ; PMID: 32534064
Secretion pathway	BcsA (inner membrane) → BcsB (periplasm) → BcsC (outer-membrane porin)	PMID: 31604608
Higher-order assembly	Hub of a megadalton Bcs macrocomplex	PMID: 29234007
Physiology in <i>P. putida</i>	Water-stress-induced; desiccation tolerance + rhizosphere colonization	PMID: 21507177
Family / domains	AcsB/BcsB family; PF03170; IPR003920/ IPR018513	UniProt Q88JL3

Evidence Base

The report draws primarily on structural, biochemical, and physiological studies of the bacterial cellulose synthase (Bcs) system. Because BcsB is highly conserved and *P. putida* Q88JL3 is an unambiguous member of the AcsB/BcsB family (PF03170), mechanistic conclusions from the well-characterized homologs transfer directly, supplemented by *P. putida*-specific physiology.

PMID	Study (organism/system)	Contribution	Relationship to findings
23222542	Crystal structure of BcsA–BcsB (<i>R. sphaeroides</i>) with translocating polysaccharide	Defines BcsB as membrane-anchored periplasmic partner of catalytic BcsA; processive single-glucose extension	Supports F1 (core identity)
24704788	Mechanism of c-di-GMP activation of cellulose synthase	Gating-loop/salt-bridge allosteric activation mechanism	Supports F2 (regulation)
29234007	Structure/assembly of the Bcs secretion system (<i>E. coli</i>)	Megadalton macrocomplex; BcsB in higher-order assembly	Supports F3 (macrocomplex)
32534064	Local c-di-GMP signaling for pEtN-cellulose (<i>E. coli</i>)	DgcC/PdeK directly bind BcsB → local c-di-GMP source & sink	Supports F3 (signaling scaffold)
27214134	Purified AcsAB complex (<i>Gluconacetobacter hansenii</i>)	UDP-glucose → β -1,4-glucan; c-di-GMP responsive; inner-membrane localization	Supports F5 (biochemistry/localization)
31604608	Cellulose synthase outer-membrane channel + TPR domain	BcsC porin secretes cellulose across outer membrane via TPR-spanned periplasm	Supports F6 (secretion path)
21507177	Cellulose/Pea EPS in <i>P. putida</i> biofilm & water stress	<i>bcs</i> minor for biofilm but aids rhizosphere colonization; strongly water-stress induced	Supports F4 (<i>P. putida</i> physiology)
35001503	CfcA/CfcR two-component system (<i>P. putida</i> KT2440)	Salt-responsive diguanylate cyclase setting c-di-GMP for biofilm	Context for F4 (upstream signaling)
37927230	Phosphodiesterase DibA–LapD (<i>P. putida</i>)	c-di-GMP network controlling biofilm via LapD	Context for F4 (upstream signaling)
38237678	Dual GGDEF/EAL enzyme PA0285 (<i>Pseudomonas</i> spp.)	Housekeeping phosphodiesterase affecting attachment/biofilm	Context for F4 (c-di-GMP homeostasis)

Supporting/comparative references. Structural comparisons across the glycosyltransferase superfamily — including plant cellulose synthase CESA (PMID: 32327535) and fungal β -1,3-glucan synthase (PMID: 41313306) — reinforce the conserved mechanism of membrane-embedded, processive glucan synthases. Studies of BcsA transmembrane mutants in *Komagataeibacter xylinus* (PMID: 31514737) further support the coupling of synthesis, translocation, and crystallization within the BcsA–BcsB machine.

Consistency of the evidence. All independent lines — X-ray structures of the BcsA–BcsB complex, biochemical reconstitution of an activity-competent AcsAB complex, EM of the megadalton assembly, and *P. putida* genetics — converge on the same model. No source in the reviewed literature contradicts the assignment of BcsB as the periplasmic accessory subunit of cellulose synthase.

Limitations and Knowledge Gaps

1. **No *P. putida*-specific structural or biochemical study of BcsB.** The mechanistic picture rests on homologs (*R. sphaeroides*, *E. coli*, *Gluconacetobacter/Komagataeibacter*). While family membership (PF03170) makes transfer of function robust, direct characterization of Q88JL3/PP_2636 has not been performed. The precise domain architecture (number of carbohydrate-binding/flavodoxin-like periplasmic domains) of the *P. putida* protein has not been experimentally verified here.
2. **BcsB's exact catalytic contribution remains non-enzymatic but mechanistically incompletely defined.** BcsB is understood as a scaffold/guide, but the extent to which it actively shapes glucan translocation versus passively housing the chain is not fully resolved for any homolog, and not at all for *P. putida*.
3. **Modification state of the *P. putida* cellulose is unknown.** In *E. coli*, the product is chemically modified (phosphoethanolamine-cellulose, pEtN-cellulose). Whether *P. putida* KT2440 produces plain cellulose or a modified variant, and the identity of the accessory subunits (e.g., BcsE/F/G, BcsZ) in its operon, were not established in this investigation.
4. **Local signaling scaffolding not confirmed in *P. putida*.** The DgcC/PdeK–BcsB interaction (source/sink microdomain) is demonstrated in *E. coli*. The specific diguanylate cyclase/phosphodiesterase that couples to BcsB in *P. putida* KT2440 is not identified, though the organism's rich c-di-GMP network (CfcR, DibA, BifA, PA0285 ortholog, etc.) provides candidate regulators.

5. **Quantitative phenotypic weight.** The contribution of cellulose to *P. putida* desiccation tolerance and rhizosphere fitness was shown qualitatively (competition assay, expression induction) but effect sizes and the interplay with the dominant Pea exopolysaccharide were not quantified in this report.

Proposed Follow-up Experiments / Actions

1. **Verify domain architecture of Q88JL3 computationally.** Run structure prediction (AlphaFold) and domain parsing on PP_2636 to confirm the membrane anchor plus periplasmic carbohydrate-binding domains, and superpose against the *R. sphaeroides* BcsB (PDB [23222542](#)) from [PDB](#) to confirm the fold.
 2. **Annotate the *P. putida* KT2440 *bcs* operon.** Map genes flanking PP_2636 to identify BcsA, BcsC, and accessory subunits (BcsE/F/G/Z) and any adjacent diguanylate cyclase/phosphodiesterase genes that might form a BcsB-centered signaling module analogous to DgcC/PdeK.
 3. **Construct and phenotype a clean $\Delta bcsB$ (PP_2636) mutant.** Assay cellulose production (Calcofluor/Congo red binding), biofilm architecture, desiccation survival, and rhizosphere competitive colonization versus wild type and Δpea , to quantify BcsB's specific contribution.
 4. **Test c-di-GMP responsiveness in *P. putida*.** Modulate intracellular c-di-GMP (e.g., via CfcR overexpression or DibA/BifA manipulation) and measure cellulose output, testing whether the *P. putida* synthase is allosterically gated as in homologs.
 5. **Probe BcsB protein interactions in *P. putida*.** Use co-immunoprecipitation or bacterial two-hybrid assays to test whether a specific DGC/PDE binds BcsB to establish a local c-di-GMP microdomain, mirroring the *E. coli* DgcC/PdeK–BcsB paradigm.
 6. **Characterize the cellulose product chemistry.** Determine whether *P. putida* KT2440 secretes unmodified cellulose or a modified (e.g., pEtN) form, using compositional/NMR analysis of purified exopolysaccharide.
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Conclusion

bcsB (Q88JL3, PP_2636) encodes **BcsB, the membrane-anchored, periplasmic accessory subunit of bacterial cellulose synthase** in *P. putida* KT2440. It is non-catalytic but forms an obligate inner-membrane complex with the glycosyltransferase BcsA, which polymerizes **UDP-glucose into β -1,4-glucan (cellulose)** and translocates it across the inner membrane. BcsB's periplasmic domains guide the nascent chain from BcsA toward the outer-membrane **BcsC** porin, and BcsB additionally scaffolds a megadalton Bcs machine and a local c-di-GMP source/sink that allosterically switches the synthase on. In *P. putida*, the resulting cellulose is a **water-stress-induced biofilm matrix polymer that promotes desiccation tolerance and rhizosphere colonization**.

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